

Penguins

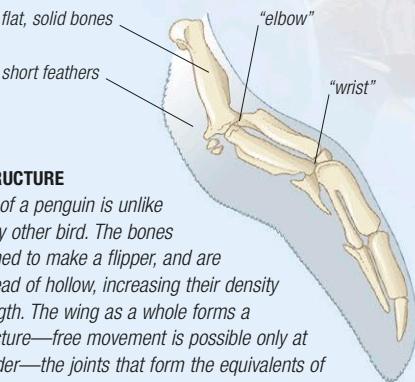
PHYLUM	Chordata
CLASS	Aves
ORDER	Sphenisciformes
FAMILIES	1 (Spheniscidae)
SPECIES	18

These distinctive flightless seabirds are adapted for swimming and surviving life in extreme cold. Penguins spend most of their lives in water, propelling themselves with flipperlike wings in pursuit of fish, krill, and squid. They have a thick coat

of short, stiff, overlapping feathers that streamlines the body, repels water, and conserves heat. Most species come ashore during the warmer months to breed, usually forming large colonies. On land, they do not perch but instead stand or glide over ice and move with a waddling gait. Penguins are confined to the seas of the Southern Hemisphere. Although most common in cold climates, several species are also found where cold currents flow north into tropical regions. Only 2 species, the emperor and Adelie penguins, spend winter in Antarctica.

Anatomy

Penguins have a plump body with short legs and webbed feet that are set so far back that they must stand upright on land, balancing on their feet and short, stiff tail. When walking, they put their weight on the soles of their feet, hence their awkward gait. On snow or ice, they may toboggan on their bellies, using their feet and flippers for propulsion. A penguin's body is streamlined in water, and is covered with extremely short feathers that form a sleek, friction-free surface. The wings are specially flattened into flippers (see below). When underwater, they use their feet and tail as a rudder. Penguins have 3 layers that provide waterproofing and insulation: a dense mass of overlapping, oil-tipped feathers; a thick layer of fat under the skin; and, in between, a layer of air that is warmed by the body. The plumage of penguins is black or gray above and white below. Any coloration or ornamentation (such as crests and eye tufts) is confined to the head and neck.



WING STRUCTURE
The wing of a penguin is unlike that of any other bird. The bones are flattened to make a flipper, and are solid instead of hollow, increasing their density and strength. The wing as a whole forms a rigid structure—free movement is possible only at the shoulder—the joints that form the equivalents of the wrist and elbow in humans being relatively inflexible. Well-developed muscles, as large as those of flying birds, help power the flapping of the wings.

COMMUNICATION IN COLONIES
Apart from the yellow-eyed penguin, all penguins (such as these king penguins) form colonies. When gathered in large numbers, they use calls and visual displays to locate their mates and young.

Colonies

Most penguins breed in colonies, which can consist of hundreds of thousands of birds. Nests are made of grass, feathers, or pebbles, and the female lays either one or 2 eggs. In some species, once an egg is laid, the female leaves the nest to feed, while the male keeps the egg warm, holding it on top of his feet and beneath folds of skin on his belly. Penguins are still able to walk when the egg is in this position. The first period of incubation can last weeks or even months, during which time the male does not feed but lives on stored fat. When the female returns, the pair take turns to guard the eggs while the off-duty bird feeds at sea.

Swimming

Penguins use 3 different swimming techniques. When idling, they swim slowly at the surface, paddling with their wings, and with their head and tail raised. When hunting, they dive below the surface and effectively fly underwater, flapping their wings to provide power. Most dives last about a minute, but dives of 20 minutes have been recorded. The third form of movement is called porpoising, in which penguins swimming near the surface periodically leap out of the water to breathe.



DIVING
Penguins (here, a king penguin) move much more efficiently in water than they do on land. Some species can swim at speeds of 9 mph (14 kph).



BURROWING
Not all penguins nest above ground. In open terrain, Magellanic penguins nest in shallow burrows to protect themselves from the elements and from predators, including mammals and other birds.

HUDDLING
In harsh conditions, emperor penguins and chicks huddle together for warmth. The center of a huddle can be about 18°F (10°C) warmer than the edges. Birds take turns to occupy the outermost positions.

